

Fig. 26.5. John Wheeler in 1971, lecturing about singularities, black holes, and the universe.

From that private lecture, I emerged a convert. From Wheeler's public lectures and writings, many other physicists emerged as converts and embarked on a quest to understand singularities and their quantum gravity laws. That quest continues today. That quest produced superstring theory, which in turn led to a belief that our universe must be a brane residing in a higher dimensional bulk (Chapter 21).

Naked Singularities?

It would be fabulous if we could find or make a singularity *outside* a black hole. A singularity not hidden beneath a black hole's event horizon. A *naked singularity*. Then in *Interstellar* the Professor's task could be easy. He might extract the crucial quantum data from a naked singularity in his NASA lab.

In 1991, John Preskill and I made a bet about naked singularities with our friend Stephen Hawking. Preskill, a Caltech professor, is one of the world's great experts on quantum information. Stephen is the